

# Hyunwoo Yang

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## EDUCATION

### DigiPen Institute of Technology

2026 – Present  
(Expected 2028)

*B.S. in Computer Science, Real-Time Interactive Simulation | GPA: 3.913 | Redmond, Washington*

**Relevant Courses:** Computer Graphics, C/C++, Operating Systems, Data Structures and Algorithms, Databases, Network Programming

## AWARDS

- **Departmental Valedictorian** — awarded full-tuition merit scholarship
- **Korea–U.S. Advanced Technology Scholar** — selected for a national program supporting promising science and engineering students, receiving up to \$18,000 in funding

## TECHNICAL SKILLS

**Languages:** C/C++, C#, JavaScript, Assembly, MYSQL, HTML

**Other & Tools:** Multithreading and Concurrency(thread pools, job systems, mutexes, condition variables), Networking, Vulkan, OpenGL, Git, Github, Perforce, Linux, Unity, Unreal Engine,

## PROJECTS

### Trolly Trolls | Tech Lead

Sep 2026 - Present

- A physics based shopping cart battle royale game built by a team of 5 supporting up to 4 player local multiplayer where I implemented the cart's physics and movement logic

### MathCalculator | Solo Developer

Jun – Sep 2026

- Full-stack bilingual (ko/en) math web app built with React 19, Vite, Three.js and mathjs, offering 18 solvers across matrix, calculus, vector and 3D graphics, each returning step-by-step derivations rendered as LaTeX through KaTeX.
- Built the computation layer on mathjs and nerdamer covering determinants, inverses, LU/SVD decomposition, eigenvalues, derivatives, integrals, limits and Taylor series, emitting structured solution steps rather than a bare answer.
- Implemented real-time Three.js visualization: MVP matrix transforms with three light types and shadow mapping (Basic/PCF/PCF Soft), vector cross products and parametric surface meshes for partial derivatives.
- Cut initial JS payload 83% (504 KB to 85 KB gzip) with React.lazy route-level code splitting, deferring Three.js, mathjs and KaTeX to the routes that need them. Deployed as a static build on GitHub Pages.

### LostGuns | Solo Developer

01/2026 – 03/2026

- Recreated the core mechanics of the mobile game LostGuns as a 2D multiplayer shooter in Unity, built from scratch by studying gameplay footage.
- Reverse-engineered four core systems — movement, jumping, ladder traversal and weapon switching and added friend-to-friend online play for up to 8 players via Unity Relay.
- Integrated 2D assets by rigging them with DragonBones and manually authoring skeletal animations to faithfully replicate the original game's visual dynamics.
- Implemented server-authoritative hit detection with lag compensation, rewinding player positions to account for RTT and interpolation delay, clamped to a 0.4s maximum.

### I.G.Y | Tech Lead

09/2025 – 06/2026

- Led technical architecture and version control workflow for a 5-person team, translating gameplay design goals into an engine roadmap built on custom C++/OpenGL.
- Replaced the immediate-mode renderer with a GPU-instanced batch renderer, cutting draw calls  $9,256 \rightarrow 11$  and frame time  $58.6 \rightarrow 38.3\text{ms}$  ( $17 \rightarrow 26\text{FPS}$ ), by packing per-quad transform, UV and tint data into a 96-byte instance record batched across 64 texture slots.
- Expanded a Computer Graphics coursework renderer into a standalone engine, Dokev, abstracting raw OpenGL into single-function calls so teammates new to graphics could focus on gameplay and AI.
- Designed and implemented A\* pathfinding with spatial partitioning to navigate up to 500 citizen agents in real time and applied backtracking to solve constraint-based placement problems in the city simulation.
- Profiled the pathfinder and found 91% of node-reset work was wasted (1.93M nodes cleared to expand 160K). Rewrote it with generation-stamped lazy initialization and stack-allocated neighbor buffers, cutting per-node cost  $1.6\mu\text{s} \rightarrow 0.84\mu\text{s}$  and a measured 179ms frame spike to a projected 111ms.

## EXPERIENCE

### Military Service | Squad Leader (Sgt.) | Daegu, South Korea

08/2023 – 02/2025

- Led and trained an 8-person mortar squad, coordinating operations and communicating clearly under pressure to align the team and make time-sensitive decisions together.